

AutoPos UWB Field Evaluation Report

OptiTrack-Isolated Static, Dynamic, Delay, Filtering, and Robustness Analysis

FULL Erlangen 2026-05-28 analysis package

Generated from the corrected FULL 4-way comparison and reviewer-audit outputs,
2026-06-04

Abstract

This report summarizes the corrected FULL AutoPos/Vicon analysis for the Erlangen 2026-05-28 dataset. The evaluation separates anchor-layout accuracy, residual delay/layout coupling, static tag positioning, dynamic ROTO trajectory tracking, post-solve filtering, pseudo-IMU oracle fusion, and robustness diagnostics. The main deployment headline is the production mean-aggregated static point after the real production export path was switched to the T4 tag solver: **72.7 mm median, 171.5 mm P95, and 109.8 mm RMSE** for original FULL v4-io/T4. The older **69.7/173.9 mm** result is retained only as a median-estimator ablation, not as the deployed production statistic. Dynamic ROTO absolute trajectory error remains about **105.8 mm median and 231.8 mm track-level P95** on the same v4-io/T4 solver line, showing that dynamic error does not collapse simply by improving anchor geometry. The FULL-US rerun represents the BioSpur AutoPos deployment system with UWB plus the US30/F-G-H height gauge. It does not materially change the legacy 3D anchor-locked accuracy numbers; instead, it fixes the vertical gauge so that the coordinate frame is physically interpretable. Under the new height-preserving evaluation, FULL-US v4-io gives **72.5 mm median, 284.8 mm P95, and 139.1 mm RMSE** for static tag points against OptiTrack.

1 One-Page Result Summary

Table 1: Primary results. Static production uses mean aggregation per static position. ROTO values are track-median summaries over 34 tag tracks from 17 rotating captures.

Domain	Case	Median/P50	P95	RMSE or key diagnostic
Static deployed headline	FULL pro- duction mean- aggregated v4-io/T4	72.7	171.5	109.8 RMSE
Static deployment gauge	FULL- US UWB+US30 height- preserving v4-io	72.5	284.8	139.1 RMSE
Anchor deployment gauge	FULL- US UWB+US30 height- preserving v4-io	98.4	162.1	109.7 RMSE
Static legacy production	FULL pro- duction mean- aggregated v4-io/T1	74.0	282.1	139.6 RMSE
Static estimator ablation	FULL median- estimator v4-io/T4	69.7	173.9	108.9 RMSE
Known-anchor control	Vicon anchors + re- estimated delay, T4	64.1	128.4	77.7 RMSE
One-baseline control	E-H one- baseline + re- estimated delay, T4	58.1	130.2	78.4 RMSE
ROTO baseline	FULL original v4-io/T4	105.8	231.8	141.3 sample RMSE
ROTO known-anchor control	Vicon anchors + re- estimated delay, T4	105.6	200.4	125.4 sample RMSE
ROTO fixed-lag filter	FULL original T4+F4	86.3	158.2	diagnostic latency smoother

The most important interpretation is that the system has two different regimes: static tag localization benefits strongly from correct layout/delay handling and from static dwell aggregation, while ROTO remains a roughly 10 cm absolute dynamic trajectory problem even under known-anchor controls. The dynamic residual is not mainly a constant time-offset problem, not mainly the 0.8 ms protocol broadcast window, and not mainly a global anchor-layout scale problem.

2 FULL-US: UWB Plus US30 Height Gauge

The original FULL analysis and the FULL-US rerun answer slightly different questions. Original FULL `v4-io` is the UWB AutoPos self-calibration with the historical soft two-layer vertical prior. FULL-US `v4-io` is the same AutoPos pipeline after applying the US30/F-G-H ultrasound height gauge to fix the vertical coordinate gauge. OptiTrack remains the external ground truth.

Table 2: FULL, FULL-US, and OptiTrack compared. Values are 3D errors in mm against OptiTrack where applicable.

System	Coordinate mean- ing	Static tag P50/P95	Anchor P50/RMSE	ROTO P50/P95
FULL v4-io	UWB 1.0/282.1 legacy production Au- to- Pos with soft two- layer Z prior; gauge is not a di- rectly de- ployed height frame		92.8/105.4	105.8/231.8
FULL-US v4-io	UWB 2.5/284.8 height-preserving Au- to- Pos plus US30/F- G- H height gauge; ver- ti- cal gauge is deployment- interpretable		98.4/109.7 height-preserving	105.8/231.8
OptiTrack	Optical ground truth ref- er- ence	reference	0.28 mm coordinate-SD median	reference

The important result is that FULL-US is not a large accuracy-improvement module under the old metric. In the no-US versus US comparison, 8 of 10 legacy anchor-locked rows are numerically unchanged, and the remaining two change by less than 0.12 mm in P95. This is expected: a full 3D anchor/capture alignment can absorb a gauge-only coordinate-frame change.

Therefore the US30 contribution should be worded as follows. It fixes the BioSpur AutoPos coordinate gauge, especially the vertical axis, so that the output coordinate system is physically interpretable in deployment. It does not by itself make UWB ranging substantially more accurate. The correct deployment diagnostic is the height-preserving check: only a horizontal 2D rigid transform plus a vertical shift is allowed, so arbitrary 3D pitch/roll cannot erase the US height gauge. Under this check, FULL-US v4-io has static tag error 72.5/284.8 mm with 139.1 mm RMSE, and anchor layout error 98.4/162.1 mm with 109.7 mm RMSE.

3 Dataset, Reference Frames, and Evaluation Protocol

The dataset is the corrected FULL OptiTrack export from Erlangen on 2026-05-28. The main infrastructure configuration is AutoPos v4-io, evaluated against OptiTrack/Vicon anchor and tag truth. The analysis compares four major paths:

- original FULL self-calibrated AutoPos layout with solver residual corrections;
- known Vicon/OptiTrack anchor coordinates with residual delay re-estimated on that frame;
- full Sim(3) scale-to-Vicon layout with residual delay re-estimated;
- one-baseline E-H scale correction with residual delay re-estimated.

Static results summarize 24 fixed tag positions. ROTO results summarize 34 tag tracks from 17 rotating captures. Static production results use the actual production export path: every static frame is solved, then the final static point is the mean of the solved coordinates in `position_summary`. The production path was switched from the legacy T1 analytic solver to the T4 C-solver, while mean aggregation was intentionally kept unchanged. This real production run reproduced the probe result within numerical noise:

72.691/171.493/109.843 mm real run

versus

72.689/171.497/109.845 mm probe.

4 Metric Definitions

- **Static median/P95/RMSE:** 3D error between the estimated static tag point and the corrected OptiTrack tag truth.
- **ROTO ATE:** absolute trajectory error between solved UWB antenna samples and interpolated OptiTrack antenna markers.
- **ROTO track-median:** the median of per-track summary values, used to avoid over-weighting longer tracks.
- **RPE:** frame-to-frame relative position error, used as a local motion consistency metric.
- **SE(3) layout error:** rigid/reflection-normalized anchor error against OptiTrack layout. It preserves global scale bias.

- **Sim(3) layout error:** similarity residual after allowing one global scale factor. It measures non-uniform shape distortion after removing global scale.
- **Delay correction:** all `d_anchor_mm` and tag-delay terms in this report are layout-level residual software corrections fitted on top of the firmware antenna-delay setting 16436 DTU. They are not quoted as physical antenna delays.

5 Anchor Self-Positioning Accuracy

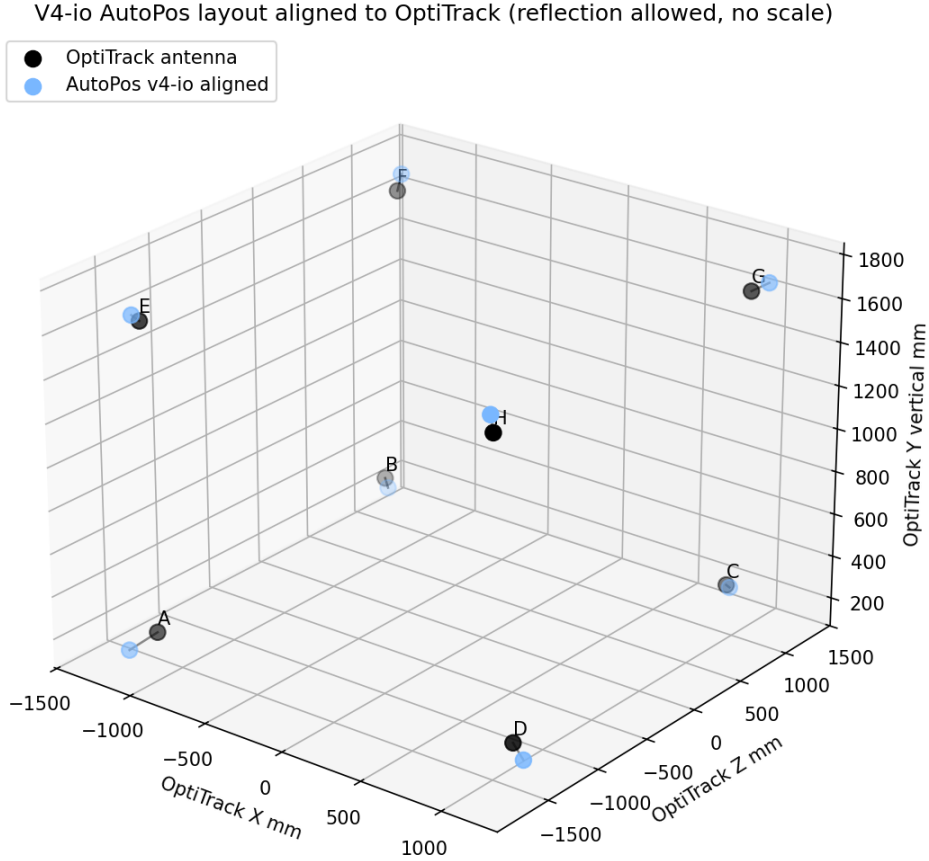


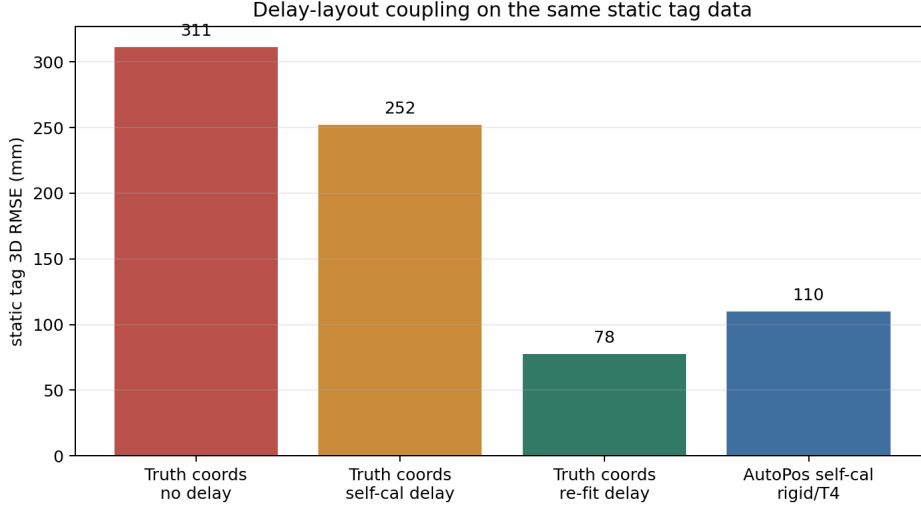
Figure 1: OptiTrack anchor layout versus AutoPos layout. The key point is not only the 3D anchor coordinate error but also the global scale shrinkage visible after alignment.

Table 3: Anchor layout absolute accuracy. SE(3) preserves scale bias; Sim(3) removes a single global scale degree of freedom.

Method	SE(3) RMSE	Median	P95	Scale bias	Sim(3) RMSE
AutoPos v1-old	101.3	84.6	154.1	-4.50%	50.1
AutoPos v4-io	105.4	92.8	156.9	-4.17%	67.1
AutoPos v2	136.5	127.8	184.0	-6.15%	60.3
AutoPos v3-lite	136.6	127.9	184.2	-6.15%	60.9
AutoPos v3-full	143.4	118.9	211.7	-3.61%	125.3

The production layout `v4-io` has a frame-normalized rigid $SE(3)$ RMSE of 105.4 mm and a $Sim(3)$ residual of 67.1 mm. The $Sim(3)$ scale is 0.958267, meaning the AutoPos layout is globally about 4.17% smaller than the OptiTrack layout. Therefore the gap between rigid and similarity RMS is not a computation bug; it is the scale degree of freedom doing real work. In paper text, the similarity number should be marked as diagnostic rather than as deployed field accuracy.

6 Delay-Layout Coupling



Truth coordinates only are insufficient; residual delay corrections must be re-estimated on the imposed coordinate frame.

Figure 2: Delay-layout coupling triangle. Imposing optical anchor coordinates is not enough; residual delay must be re-estimated on the imposed coordinate frame.

Table 4: Same static tag data under different layout and delay treatments.

Case	Interpretation	RMSE	P95
Vicon truth, no delay correction	Optical geometry alone is insufficient	311.3	453.4
Vicon truth, transplanted self-cal delay	AutoPos residual corrections are not portable	252.2	394.6
Vicon truth, re-estimated delay	Known-anchor + re-estimated-delay control	77.7	128.4
AutoPos v4-io, co-fitted delay	Self-cal geometry and delay are jointly conditioned	108.9	174.1

This is the central contribution of the OptiTrack-isolated analysis. Known anchor coordinates by themselves do not solve the tag problem; the endpoint residual delay model must be re-estimated in the same layout frame. The mechanism is also visible at the per-anchor level: WHY #9 regresses self-minus-Vicon `anchor_main_rel_A` on `v4-io` radial layout error and obtains $R^2 = 0.998$ with slope -0.982 mm/mm . In other words, the self-cal residual delay term absorbs radial coordinate and scale error almost one-to-one. This is not a solver sign bug; it is gauge absorption.

7 Static Tag Positioning

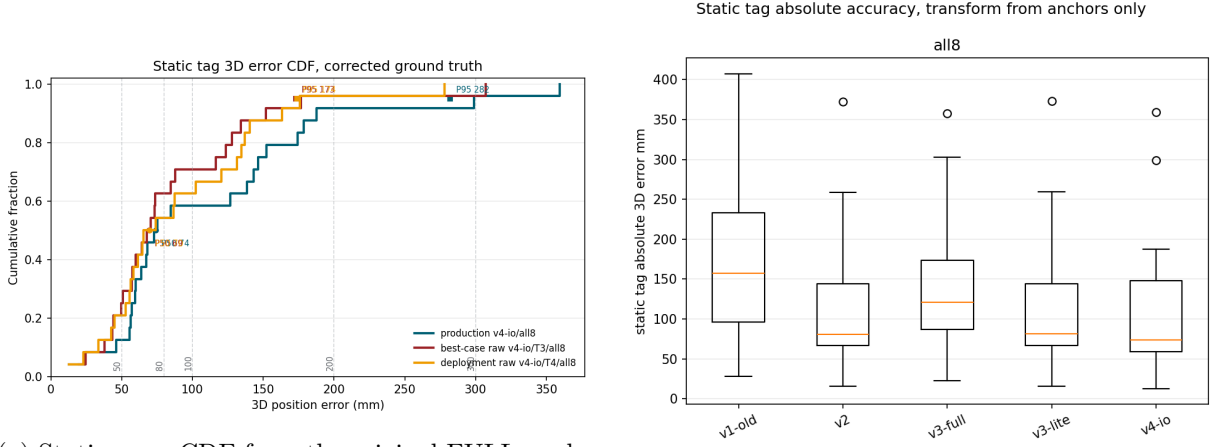


Figure 3: Real production export path after switching the frame solver to T4 while keeping mean aggregation unchanged.

Table 5: Static tag accuracy under the corrected reporting convention.

Case	Role	Median	P95	RMSE
Production mean v4-io/T4	deployed static headline	72.7	171.5	109.8
Legacy production mean v4-io/T1	old exported production path	74.0	282.1	139.6
Median-estimator v4-io/T4	estimator-sensitivity ablation	69.7	173.9	108.9
Median-estimator v4-io/T3	estimator-sensitivity ablation	69.2	173.0	106.6
Vicon truth + re-estimated delay	known-anchor control	64.1	128.4	77.7
Sim(3) scale + re-estimated delay	global scale removed	67.1	132.6	81.6
One-baseline E-H + re-estimated delay	field-measurable scale control	58.1	130.2	78.4

The production-vs-replay distinction matters. The 69.7/173.9 mm row is not the deployed static point; it is a median estimator applied in the raw replay/evaluation path. The deployed export uses mean aggregation, and after the real production T4 change the headline is 72.7/171.5 mm with 109.8 mm RMSE.



(a) Static error CDF from the original FULL analysis.

(b) Static error by position.

Figure 4: Static accuracy has meaningful position dependence and tail behavior; a single 3D RMSE does not communicate the full structure of the error.

7.1 Static Filtering and Static-Lock Interpretation

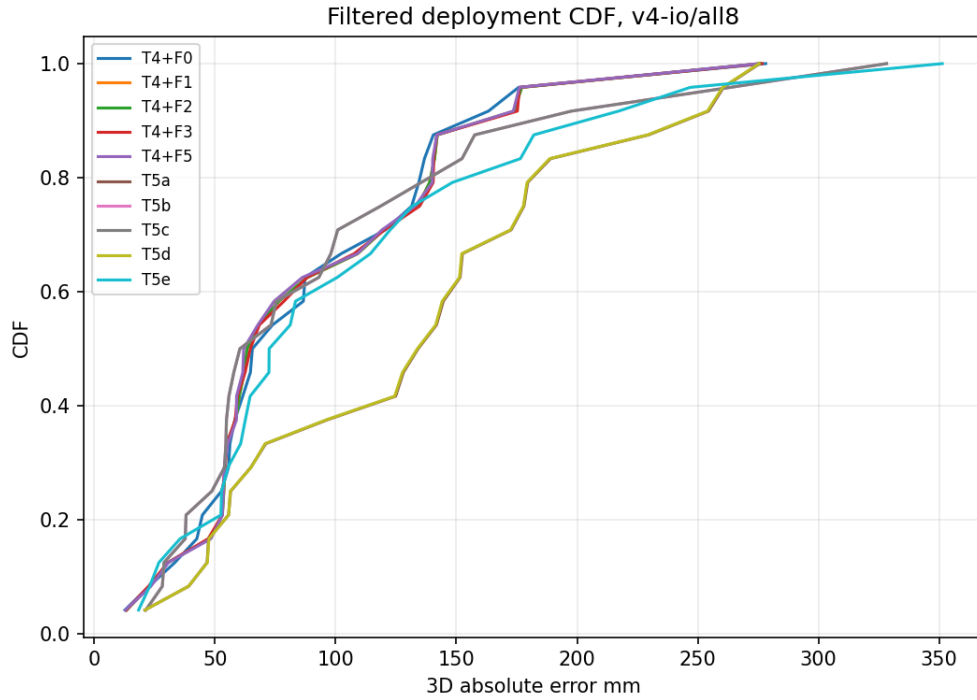


Figure 5: Static post-solve filtering. The filter reduces user-facing jitter but does not eliminate systematic layout/delay/ranging bias.

Table 6: Static stationary filter results. These are deployment/static-lock diagnostics, not raw calibration accuracy.

Solver	Median	P95	RMSE	Repeatability SD median
T4+F0 baseline	69.7	173.9	108.9	67.4
T4+F5 offline smoother	64.9	175.7	109.5	18.7
T4+F4 fixed-lag smoother	65.9	176.1	109.6	21.4
T3+F5 offline smoother	65.2	169.9	105.4	21.0

Static-lock filtering is valuable if the application knows the tag is stationary: it reduces the median per-position 3D spread from 67.4 mm to 18.7 mm. However, the P95 and RMSE remain close to the unfiltered values because systematic bias is not removed by smoothing.

8 Dynamic ROTO Absolute Trajectory Results

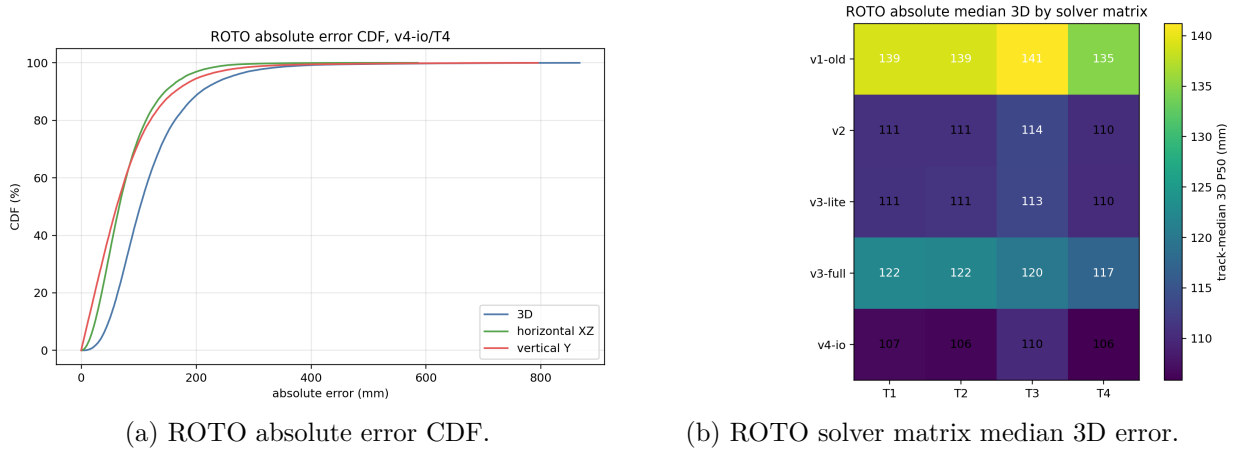


Figure 6: Dynamic ROTO accuracy remains near 10 cm median even on the same T4 solver line that performs well statically.

Table 7: ROTO absolute trajectory accuracy. P50/P95 are track-median summaries; RMSE is sample-pooled ATE RMSE.

Case	Layout/delay treatment	Track P50	Track P95	Sample RMSE
FULL original v4-io/T4	self-cal layout + solver residual corrections	105.8	231.8	141.3
Vicon anchors + delaycal	known anchors + re-estimated delay	105.6	200.4	125.4
Sim(3) scale + delaycal	global scale removed + re-estimated delay	110.5	200.7	127.8
One-baseline E-H + delaycal	one-baseline scale + re-estimated delay	106.2	200.4	126.7
Best ROTO median row	one-baseline B-C with solver delay	100.1	220.5	—

The key dynamic result is negative but important: the ROTO median does not collapse when the layout is replaced with Vicon anchors and residual delay is re-estimated. Static improves from 108.9 mm RMSE to 77.7 mm RMSE under known-anchor delaycal, but ROTO remains around 105 mm median. Therefore dynamic ROTO error includes motion, rotating-wand ranging behavior, antenna/marker geometry, and residual trajectory effects beyond static infrastructure calibration.

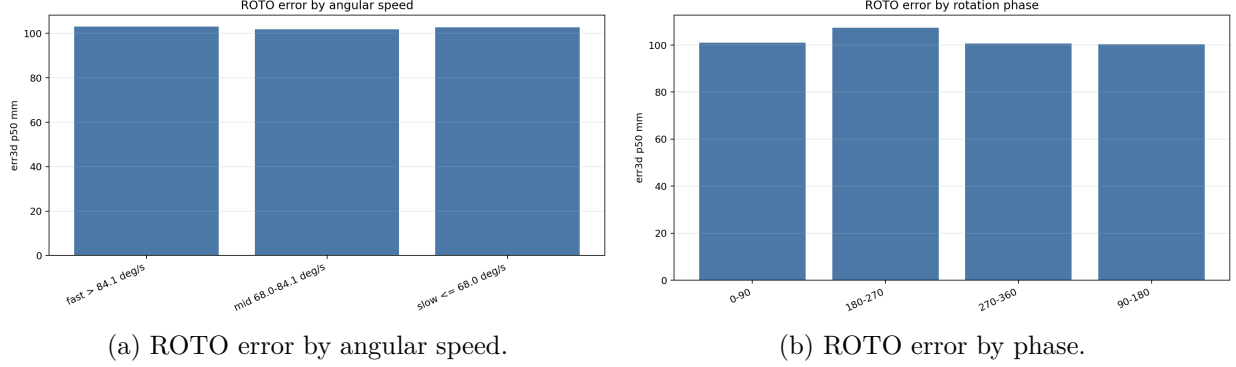


Figure 7: Speed and phase move the median only modestly; they do not explain the full dynamic residual.

8.1 Time Offset and Clock-Skew Falsification

The reviewer audit refit one constant time offset per capture, then added an affine time skew model

$$t_{\text{query}} = t_{\text{UWB}} + \beta + \alpha(t_{\text{UWB}} - t_{\text{ref}}).$$

For original FULL v4-io/T4, the sample-pooled P50 sequence was 102.63 mm at the current beta, 102.69 mm after constant-offset refit, and 102.58 mm after skew refit. The marginal skew gain was only 0.11 mm pooled P50. The median fitted skew was -112 ppm with a 228 ppm IQR, and the verdict was SKEW_EXCLUDED. A 0.8 ms protocol window corresponds to only 0.51 mm median and 0.72 mm P95 displacement at the observed ROTO speeds.

8.2 Single-Shot Static Versus ROTO

The static replay was also evaluated at the per-frame level. For self-cal v4-io/T4, static single-shot P50 was 89.6 mm and ROTO single-shot P50 was 102.6 mm, so the dynamic excess in pooled 3D P50 was about 13.1 mm. Lost static averaging explains part of the static-vs-dynamic gap, but GDOP-conditioned bins and component splits still show a limited real dynamic term. For the Vicon-delaycal control the dynamic excess is larger, about 24.9 mm P50, again supporting the conclusion that dynamic ROTO is not only a layout-calibration artifact.

8.3 Legacy ROTO Self-Consistency and New OptiTrack Center Error

The old no-groundtruth ROTO self-consistency metrics remain useful but should not be confused with absolute trajectory error. Original FULL v4-io/T4 has:

- 25.9 mm legacy radius-difference RMS;
- 13.7 mm turn-center repeatability median;
- 37.6 mm inner/outer center-separation median;
- 72.1 mm OptiTrack/Vicon turn-center absolute 3D RMS.

The absolute center RMS stays about 72.1 mm for original FULL, 72.7 mm for Vicon anchors + delaycal, 76.7 mm for full similarity scale + delaycal, and 77.1 mm for one-baseline E-H + delaycal.

9 ROTO Filtering and Pseudo-IMU Oracle

Table 8: Post-solve ROTO filtering. These filters operate on solved positions, not raw ranges.

Case	Track P50	Track P95	Sample RMSE	Center RMS
FULL F0 baseline	105.8	231.8	141.3	72.1
FULL F4 fixed lag	86.3	158.2	103.1	69.1
FULL F5 offline RTS	83.3	148.6	98.9	68.3
Vicon F0 baseline	105.6	200.4	125.4	72.7
Vicon F4 fixed lag	84.0	145.4	95.1	70.5
Vicon F5 offline RTS	82.7	139.1	91.2	70.0

Fixed-lag and offline smoothing reduce dynamic scatter substantially, but this is a trajectory-output/latency tradeoff. The calibration-level claim should keep the unfiltered ROTO result, while F4/F5 should be reported as deployment-filter or offline upper-bound diagnostics.

Table 9: Pseudo-IMU oracle replay. The motion prior is derived from OptiTrack marker motion and is lever-armed to the UWB antenna point.

Case	Fusion	Track P50	Track P95	Sample RMSE	dR RMS
FULL original	PI0 baseline	105.8	231.8	141.3	25.9
FULL original	PI1 causal oracle	66.1	97.5	71.6	12.3
FULL original	PI4 offline oracle	58.7	81.5	61.8	5.3
Vicon delaycal	PI0 baseline	105.6	200.4	125.4	18.0
Vicon delaycal	PI1 causal oracle	64.0	100.2	69.8	11.7
Vicon delaycal	PI4 offline oracle	59.9	82.0	63.1	4.3

The body-fit antenna prior residual is only 0.6 mm P50-of-P50 and 1.5 mm P50-of-P95 across tracks, so the pseudo-IMU is not accidentally using the marker body centroid as the antenna point. It is still an oracle: it uses OptiTrack-derived motion, not a real IMU. The correct paper framing is that a properly lever-armed real inertial relative-motion channel is worth implementing and validating in a raw-range EKF/UKF, not that the current system already has a deployable UWB+IMU result.

10 Robustness and Reporting Checklist

Table 10: Numerical precision and dropout stress. Bootstrap values are not independent deployment repeatability.

Case	Diagnostic	Static SD	Delay rel-A SD	Static P50/P95	Note
Original self-cal	raw-pair median bootstrap	1.02	0.56	76.3/180.9	within-campaign sampling SE
Vicon truth + delaycal	residual- delay bootstrap	0.00	0.45	65.0/126.7	coordinate fixed by control
Sim(3) scale + delaycal	raw-pair median bootstrap	0.97	0.53	69.3/134.8	within-campaign sampling SE
One-baseline E-H	raw-pair median bootstrap	1.20	0.54	60.6/127.9	within-campaign sampling SE

The raw-pair bootstrap result is intentionally not called deployment repeatability. It agrees with the analytical median sampling standard error and therefore measures large- n numerical precision inside the same campaign. Independent repeated AutoPos deployments, repeated-run delay standard deviations, PANS/manual baselines, explicit CIR/NLOS labels, and raw dynamic packet-loss re-solves remain separate evidence gaps.

Synthetic static dropout was re-solved from raw static frames. Random per-frame four-anchor stress degrades original self-cal from 76.3/180.9 mm to 87.7/292.4 mm position P50/P95, with much worse sample-level spread. Fixed random four-anchor subsets and best-GDOP four-anchor subsets make non-convergence explicit; their P50/P95 values are conditional on convergence. ROTO packet stress is solved-sample thinning, not a raw range re-solve: ATE remains stable under thinning, while RPE worsens as effective update rate drops from about 10 Hz to about 1 Hz.

11 Per-Tag and Per-Anchor Residual Structure

The WHY #9 two-way median-polish decomposition separates common-mode, tag main effects, anchor main effects, and tag-anchor interactions. Pairwise tag residual range offsets are stable across self-cal and Vicon-delaycal scenarios:

- BS2DCE minus BSF66F: 31.0 mm in self-cal, 34.6 mm in Vicon-delaycal;
- BSDC91 minus BSF66F: -12.7 mm in self-cal, -11.7 mm in Vicon-delaycal;
- BS2DCE minus BSDC91: 43.7 mm in self-cal, 46.3 mm in Vicon-delaycal.

The maximum cross-scenario change of these tag differences is only 3.6 mm, and the median interaction terms are 0.9 mm and 2.4 mm. These are clean deployable per-tag range-offset trim targets. They should not be over-named as antenna delay; they combine device antenna delay, fixed installation effects, and stable power or range-bias terms into one scalar correction.

The anchor side is different. Absolute per-anchor and per-tag delay are not identifiable from tag-to-anchor ranges alone because of global delay/scale gauge coupling. The self-cal anchor residual structure absorbs layout radial error almost one-to-one, so self-cal `d_anchor` should be reported as a layout-level residual correction, not a physical anchor antenna-delay measurement.

12 Recommended Claims

1. Use the unfiltered production mean-aggregated `v4-io/T4` result as the deployed static headline: **72.7 mm median, 171.5 mm P95, 109.8 mm RMSE**.
2. Report 69.7/173.9 mm only as a median-estimator ablation on the same solver line, not as the deployed production output.
3. Report FULL-US as the BioSpur UWB+US30 deployment-gauge variant. It should not be claimed as a major accuracy improvement over FULL under legacy anchor-locked metrics; its contribution is fixing the physically interpretable vertical gauge. Quote the height-preserving `v4-io` static diagnostic as **72.5/284.8 mm**, RMSE **139.1 mm**.
4. State the known-anchor + re-estimated-delay control as a diagnostic control, not as a plain lower bound: it is the best static control at 64.1/128.4 mm and 77.7 mm RMSE, but it requires OptiTrack anchors and re-estimated residual delay.
5. Present delay-layout coupling as a core result: the 311/252/77 mm triangle and the $R^2 = 0.998$, slope -0.982 radial-absorption mechanism are two views of the same non-separability.
6. State ROTO conservatively: unfiltered dynamic absolute trajectory error is about 10 cm median, with P95 about 20–23 cm depending on the control row.
7. Treat F4/F5 filtering and PI1/PI4 pseudo-IMU results as deployment-filter diagnostics or oracle upper bounds, not calibration accuracy.
8. Do not report only one 3D RMSE. Split repeatability, absolute error, scale bias, shape distortion, delay-layout coupling, downstream tag error, and tail behavior.

13 Limitations and Next Experiments

The present dataset covers the corrected OptiTrack-isolated FULL analysis in depth, but it still lacks several independent evidence streams:

- independent repeated AutoPos deployments or layout splits;
- true repeated-run residual-delay standard deviation;
- a practical PANS/manual baseline in the same dataset;

- explicit CIR/NLOS labels tied to each link;
- raw dynamic ROTO range re-solve under physical packet loss;
- real IMU data with calibrated IMU-to-UWB-antenna extrinsics.

The most valuable next experiment is a real UWB+IMU fusion run. The pseudo-IMU oracle shows that a correctly lever-armed relative-motion prior has enough leverage to bring ROTO from 105.8/231.8 mm to 66.1/97.5 mm in the strong causal oracle case. The deployable version should use real inertial measurements, an antenna-frame extrinsic calibration, and a raw-range EKF/UKF rather than only smoothing solved positions.

14 Source Artifacts

This LaTeX report is based on the following generated artifacts:

- `FULL_4way_comparison/reports/FULL_4WAY_BIG_COMPARISON.md`
- `FULL_4way_comparison/reviewer_audit/reports/MEASUREMENT_REVIEWER_AUDIT.md`
- `FULL_4way_comparison/reporting_checklist/reports/REPORTING_CHECKLIST_AUDIT.md`
- `FULL_4way_comparison/resilience_gap_audit/reports/RESILIENCE_GAP_AUDIT.md`
- `FULL_4way_comparison_US/reports/FULL_4WAY_BIG_COMPARISON.md`
- `FULL_NO_US_VS_US/reports/FULL_NO_US_VS_US_COMPARISON.md`
- `FULL_4way_comparison/tables/full_4way_headline_comparison.csv`
- `FULL_4way_comparison/reporting_checklist/tables/*.csv`
- `FULL_4way_comparison/reviewer_audit/tables/why*.csv`